AI-Based Smart Plant Monitoring and Care System

AI-Based Smart Plant Monitoring and Care System is a web-based application designed to assist users in monitoring plant health and providing intelligent plant care using Artificial Intelligence (AI). The system supports two major categories of plants: **Garden Plants** and **Kitchen Garden Plants**. It enables users to upload images of leaves, flowers, fruits, or entire plants for AI-powered analysis and receive accurate information about plant health. The application is designed with a simple and user-friendly interface, making it suitable for home gardeners, gardening enthusiasts, students, and beginners who wish to maintain healthy plants with minimal effort.

The system integrates multiple functionalities, including **plant identification, plant disease detection, nutrient deficiency detection, pest identification, disease progress tracking, analysis history,** and a **Plant Encyclopedia**. Using image classification and deep learning techniques, the AI model identifies the plant species, detects diseases, recognizes nutrient deficiencies such as nitrogen, phosphorus, potassium, and magnesium deficiencies, and identifies common pests affecting the plant. The system also displays the confidence score of AI predictions, along with possible symptoms, treatment methods, fertilizer recommendations, preventive measures, and general plant care guidelines to help users take appropriate action.

In addition, the Plant Encyclopedia provides detailed information about different plants, including their scientific names, ideal growing conditions, watering schedules, sunlight requirements, soil preferences, temperature conditions, and common diseases and pests. Users can also access their previous analysis reports and monitor the progress of plant recovery by comparing results over time. An Admin Module is included to manage user information, plant data, disease details, pest information, nutrient deficiency records, and treatment recommendations, ensuring that the system database remains accurate and up to date.

The proposed system aims to simplify plant monitoring and improve plant care by providing an intelligent, reliable, and easy-to-use platform. It helps users detect plant problems at an early stage, reducing the risk of severe damage and improving plant growth. By combining Artificial Intelligence with comprehensive plant care knowledge, the system offers an efficient, accurate, and accessible solution for maintaining healthy plants, supporting sustainable gardening practices, and promoting better plant management for both home gardens and kitchen gardens.